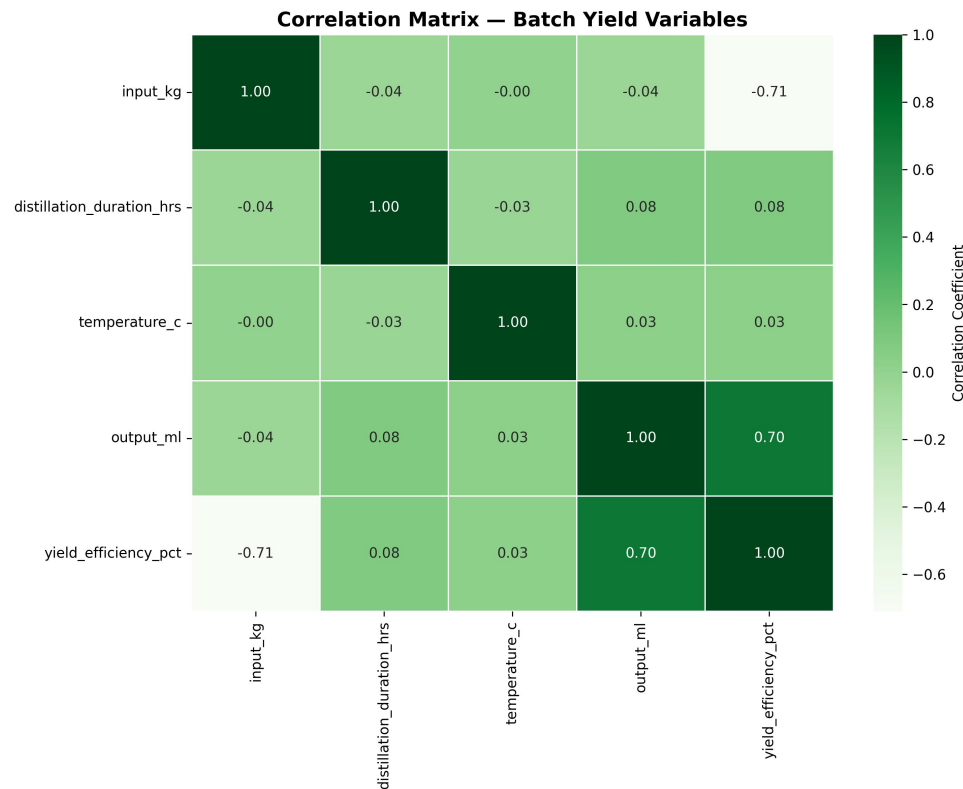


TBI-GEU Summer Internship Program 2026
Week 3 — Visual Narratives

Intern ID: 26101136 | AI-Enabled Business Intelligence | Herbal & Aromatics

Chart 1 — Bar Chart: Average Yield Efficiency by Plant Variety

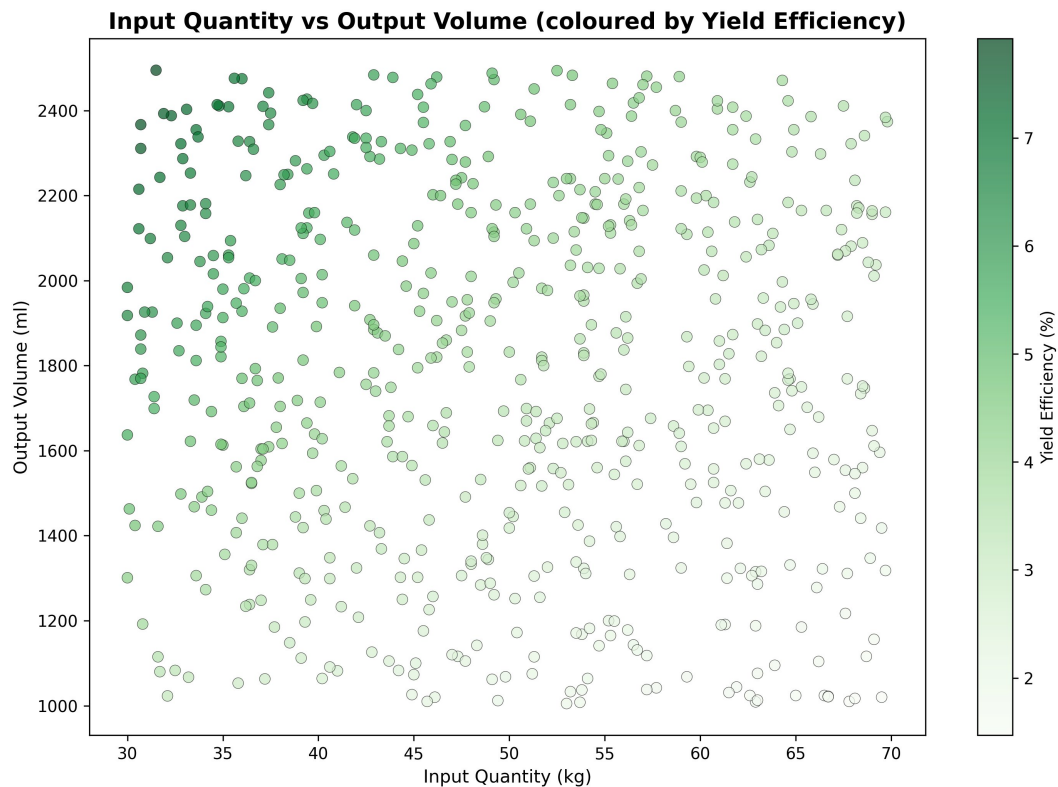


This bar chart compares the average extraction yield efficiency across the five plant varieties processed at the distillation unit: Mint, Eucalyptus, Chamomile, Lavender, and Rosemary. Each bar represents the mean yield efficiency percentage calculated across all batches of that variety, with exact values labelled above each bar for precision.

The most notable observation is how tightly clustered the results are. Mint leads marginally at 3.88%, while Rosemary trails at 3.70% — a difference of less than 0.2 percentage points across the entire variety range. No single variety dramatically outperforms the others.

For the business, this means plant variety alone is not the primary lever for improving yield. Production planning should not over-prioritise one variety over another based on yield expectations; instead, attention should shift toward process variables like input quantity and distillation conditions, which the other charts suggest have a far greater impact on efficiency outcomes.

Chart 2 — Scatter Plot: Input Quantity vs Output Volume

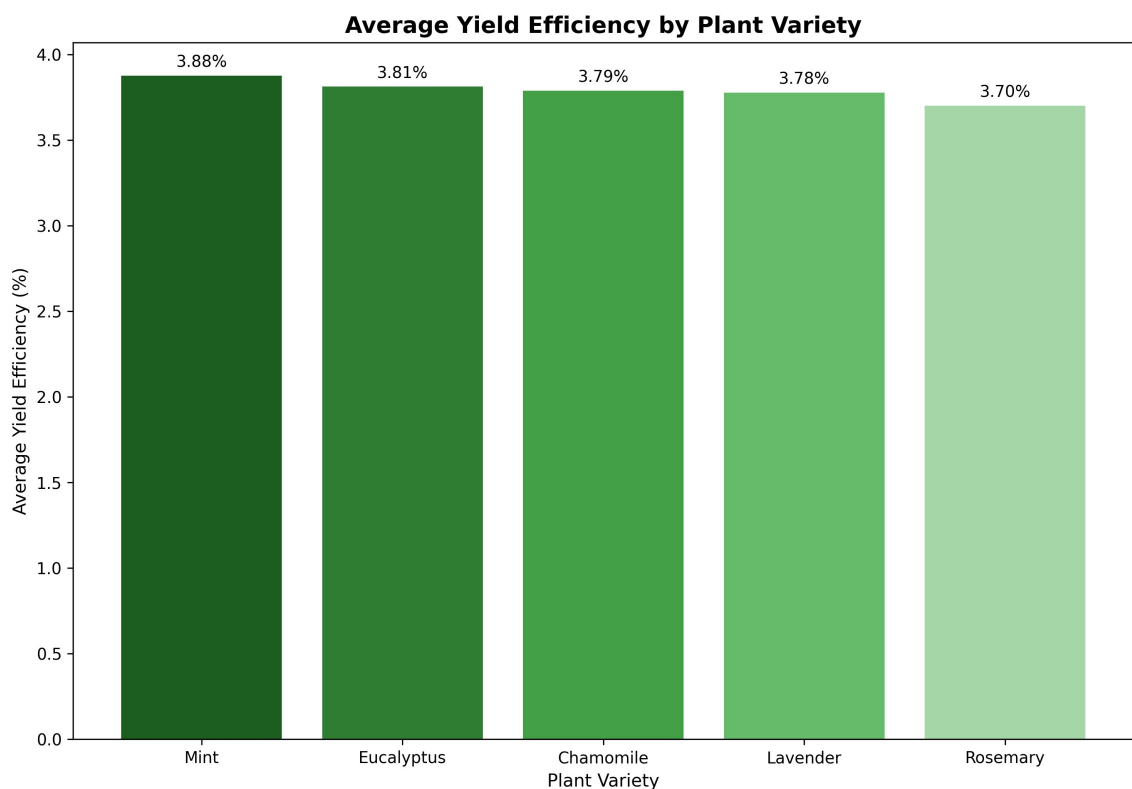


This scatter plot maps each batch's input quantity (kg) against its output volume (ml), with point colour indicating yield efficiency — darker green represents higher efficiency. This lets us see three variables simultaneously: input, output, and efficiency.

The clearest pattern is that the darkest, most efficient points cluster in the lower-left region, where input quantities are smallest (30-40kg) yet output remains high. As input quantity increases toward 60-70kg, points consistently fade to lighter shades, confirming that bigger batches do not scale efficiency proportionally.

This is a critical operational insight: processing smaller, more controlled batches appears to extract oil more efficiently than larger bulk batches. The business should consider testing a shift toward smaller batch sizes, even if it means more distillation cycles, since the yield gain per kilogram processed may outweigh the added time cost.

Chart 3 — Heatmap: Correlation Matrix of Batch Yield Variables



This heatmap displays the correlation coefficients between five key numeric variables: input quantity, distillation duration, temperature, output volume, and yield efficiency. Darker green cells indicate stronger positive correlation, helping identify which factors move together.

The standout finding is the -0.71 correlation between `input_kg` and `yield_efficiency_pct` — a strong negative relationship confirming that larger input batches consistently produce lower efficiency. In contrast, `output_ml` and `yield_efficiency_pct` show a strong positive correlation of 0.70, while distillation duration and temperature show almost no correlation with efficiency at all.

This tells the business exactly where to focus: input batch size is the dominant controllable factor affecting efficiency, while temperature and distillation duration can largely be standardised without fear of hurting yield. Future process improvements should prioritise batch-size optimisation over fine-tuning temperature or timing.